

Dominic Bair

✉ dbair@tennessee.edu 🌐 www.dbair.xyz
🐼 DothWalrus in Dominic Bair

Education

- 2022 – ··· 📖 **Ph.D. candidate, Mathematics**, University of Tennessee Knoxville (UTK).
2020 – 2022 📖 **M.S. Mathematics**, Montana State University (MSU).
2016 – 2020 📖 **B.S. Mathematics; Minor: Physics**, Montana State University.

Research Interests

- 📖 Percolation theory, geometric group theory, stochastic differential equations, machine learning, applications.

Publications

- 1 **D. Bair**, S. Jana, and Y. Qing, “First-passage percolation, non-positive curvature, and radial maps,” *Preprint*, 2025.
- 2 **D. Bair**, “DVM: A deep learning algorithm for minimizing functionals,” *Theses and Dissertations at Montana State University (MSU)*, 2022.
- 3 C. Potts et al., “MNPmApp: An image analysis tool to quantify mononuclear phagocyte distribution in mucosal tissues,” *Cytometry Part A*, vol. 101, no. 12, pp. 1012–1026, 2022.

Teaching

- M115 📖 **Statistical Reasoning**, University of Tennessee.
M125 📖 **Basic Calculus**, University of Tennessee.
M113 📖 **Mathematical Reasoning**, University of Tennessee.
M119 📖 **College Algebra**, University of Tennessee.
M171 📖 **Calculus I**, Montana State University.
M151 📖 **Precalculus**, Montana State University.
M161 📖 **Survey of Calculus**, Montana State University.
M121 📖 **College Algebra**, Montana State University.

Conferences and Workshops

- 2026 📖 **Young Geometric Group Theory XIV**, Korea Institute for Advanced Study, Seoul, South Korea.
 📖 **SLMath TGS Low Dimensions Workshop**, SLMATH, Berkeley, CA.
2025 📖 **ATMCS11**, Montana State University, Bozeman, MT.
 📖 **53rd John H. Barrett Memorial Lectures**, University of Tennessee.
2024 📖 **52nd John H. Barrett Memorial Lectures**, University of Tennessee.

Presentations

- 2026 📖 **Graphons and Random Graphs**, UTK Graduate Student Seminar.

Presentations (continued)

- 2025  **Introduction to First-Passage Percolation and Subadditive Processes, Parts I–III**, UTK Probability Seminar.
-  **First-Passage Percolation on Hyperbolic Graphs: Bi-infinite Geodesics, Radial Maps, and Infinitely Many Thick Triangles**, UTK Topology Seminar.
- 2021  **Deep Variational Methods**, MSU Poster Slam.
-  **Deep Variational Methods**, Walk the Walk Chalk the Chalk.
- 2020  **Spatial Statistics in Histology Images**, SIAM Northern States Section Student Chapters Conference.
-  **Spatial Statistics in Histology Images**, Data Science and Image Analysis Conference of the Pacific Northwest.
- 2019  **Spatial Statistics in Histology Images**, MSU Winter Research Celebration.

Awards and Academic Milestones

- 2026  **UTK Graduate Student Academic Achievement Award.**
- 2025  **UTK Oral Specialty Exam**, passed.
-  **UTK Preliminary Exam in Topology**, passed.
-  **UTK Preliminary Exam in Stochastics**, passed.
- 2020  **NSF GRFP Reviews**, Excellent, Excellent, Very Good. Fellowship not received.
-  **NSF and AWM Travel Grant**, to present at the Data Science and Image Analysis Conference of the Pacific Northwest.
- 2019–2020  **INBRE Grant**, to support Spatial Statistics in Histology Images research; NIH funded, MSU based grant.
- 2016–2020  **MSU Premier Scholarship**, for academic promise, leadership experience, and other honors.
- 2019  **Superior Achievement in the Field of Mathematics**, recognition by Pi Mu Epsilon.
- 2016  **Eagle Scout.**

Reading Courses

- 2026  **Fundamentals of Stein's Method (Ross)**, University of Tennessee.
- 2023  **Monte Carlo Strategies in Scientific Computing (Liu)**, University of Tennessee.

Employment

- 2022–...  **Graduate Teaching Assistant**, University of Tennessee Department of Mathematics.
- 2020–2022  **Graduate Teaching Assistant**, Montana State University Department of Mathematical Sciences.
- 2019–2020  **Undergraduate Research Assistant**, Montana State University Department of Mathematical Sciences.
- 2018–2019  **Undergraduate Research Assistant**, Montana State University Department of Physics.
- 2023–2024  **Climbing Wall Attendant/Setter**, UTK RecSports.
- 2015–2022  **Laborer**, 3 Brothers Plumbing and Heating.
- 2020  **Laborer**, Peak Property Management.

Service and Certifications

- 2023  **CPR and First Aid**, American Red Cross.
- 2018–2020  **Assistant**, National Center for Unwanted Firearms.
- 2015–2017  **Firefighter**, Clancy Volunteer Fire Department.

Skills

- Coding/Software  MATLAB, Python, Julia, $\text{\LaTeX}$, Windows, Linux.
- Languages  German (A1).